

NYC Airbnb Listings — Problem Statement

Academic Data Analytics Project (Power BI Dashboard)

Background

Airbnb has grown into one of the largest short-term rental marketplaces in the world, with hundreds of thousands of listings across major cities. New York City alone hosts over 100,000 active and historical listings spanning five boroughs, multiple room types, and a wide range of hosts — from individual owners to large-scale corporate property managers. This scale generates rich but messy data: inconsistent formatting, missing values, duplicate records, and fields that mix text and numeric information (e.g., prices stored as "\$966"). Raw data in this state cannot be directly analyzed or visualized, and any business decisions drawn from it without proper cleaning risk being misleading.

Problem Statement

The raw NYC Airbnb dataset (102,599 listings, 26 attributes) contains significant data quality issues — 541 duplicate records, columns with near-total missingness (e.g., "license" missing in 99.998% of rows), inconsistent column naming, and price/fee fields stored as unstructured text — that prevent it from being used directly for analysis or reporting. Additionally, stakeholders (e.g., prospective hosts, market analysts, or platform managers) lack a consolidated, visual way to understand how listings are distributed across boroughs and neighbourhoods, how pricing and availability vary by room type and location, how guest engagement (reviews) behaves across the market, and how host behavior (verification, cancellation policy, portfolio size) shapes the marketplace.

The goal of this project is to clean and structure the raw dataset into an analysis-ready form, and to design an interactive Power BI dashboard that surfaces actionable insights on pricing, geography, host behavior, and guest engagement — enabling faster, data-driven understanding of the NYC short-term rental market.

Key Questions the Project Answers

- How are listings distributed across NYC boroughs and neighbourhoods, and by room type?
- How does pricing vary (or fail to vary) across location, room type, and other listing attributes?
- Which neighbourhoods have the highest concentration of listings, and how do they compare on price?
- How engaged are guests with listings, as measured by review volume and review ratings?
- How do host characteristics — identity verification, cancellation policy, portfolio size — vary across the market?
- How does listing availability differ across boroughs and room types, and what does that suggest about demand?
- What data quality issues exist in the raw dataset, and how do they need to be addressed before analysis?

Scope & Deliverables

- A cleaned, analysis-ready dataset (duplicates removed, data types corrected, columns standardized).
- An interactive Power BI dashboard (.pbit) visualizing pricing, geography, host, and engagement metrics.
- A documented set of business insights and data-quality findings suitable for a portfolio and interview discussion.